

ROBOT CAR GUIDE

The Robot Car provides a mid-size platform for a wide range of robotics projects and experiments. The start kit supports the mechanical frame and electrical components to achieve the mission of tracking line for the robot. It covers the basic principles of analog and digital circuits as well as robot sensing and control mechanisms.

Features

- 6 Digital IR Transceivers
- 2 independent DC motors
- Support dual full bridge motor driver
- Multiple mounting locations for sensors, add-ons, etc.
- Implement of control logical design



Bill of Materials

Item No.	Quantity	Accessories	Description
(1)	1	Breadboard	PB-103, 2390 tie-points, non-powered
(2)	2	Chassis Plate	Top and bottom parts of Robot Car main frame
(3)	2	Motor Mount Plate	For mounting DC motor
(4)	1	Wire Stripper	HT-1042, strips small wires
(5)	2	DC Motor	Gear Motor, 12VDC, 200 RPM, 4mm shaft
(6)	2	Wheel	Soft plastic model wheel
(7)	1	Battery Monitor	For connecting battery and breadboard
(8)	1	Battery Charger	For charging battery
(9)	1	Battery	12V
(9)	1	Medium Tool Box	F-200, storing Robot Car body parts
(11)	1	Transceiver Mount Bar	For mounting digital IR transceiver

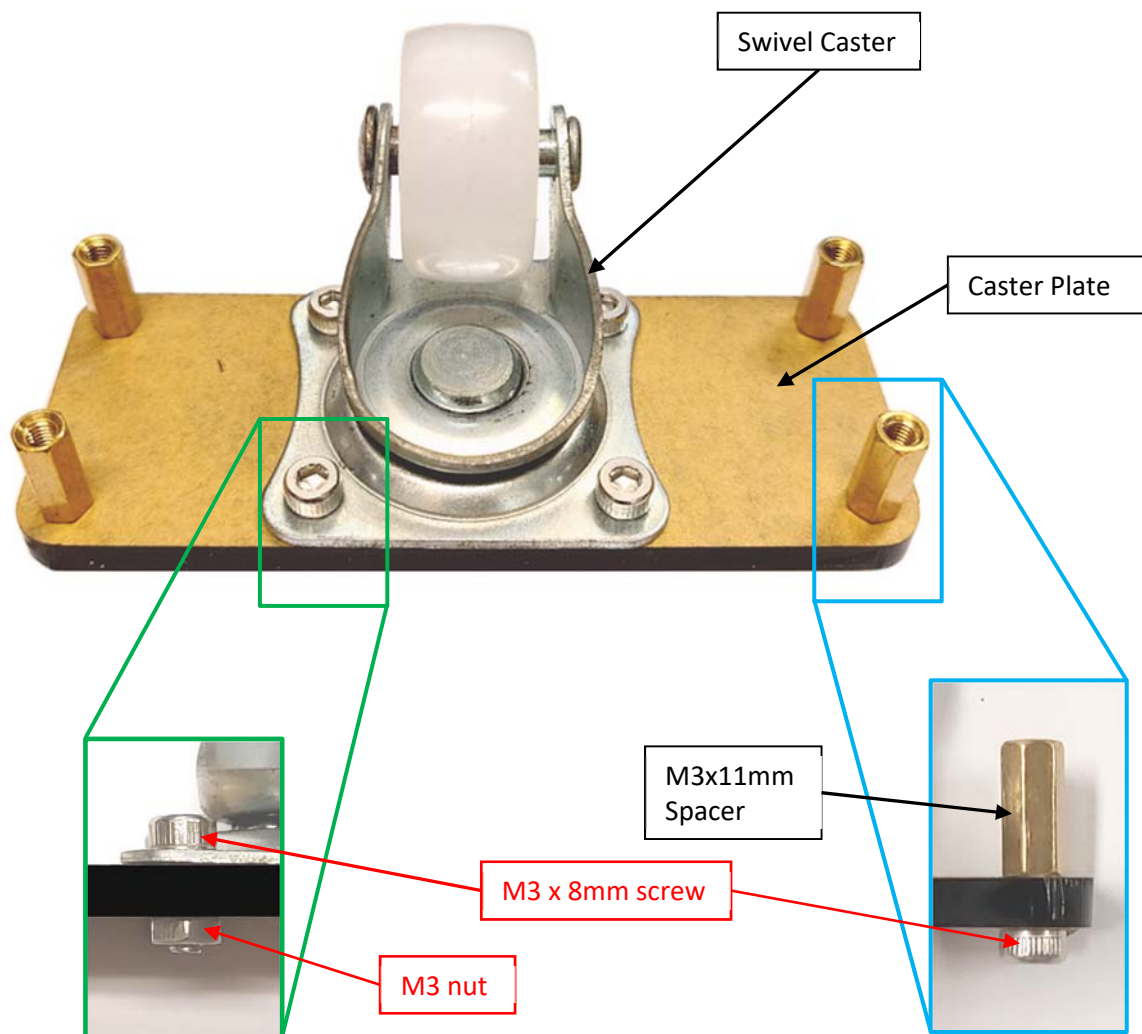
(12)	2	Bar, 6 holes	Parts of Robot Car main frame
(13)	2	Bar, 11 holes	Parts of Robot Car main frame
(14)	1	Caster Plate	For mounting swivel caster
(15)	4	Chassis Stand	Parts of Robot Car main frame
(16)	1	Swivel Caster	Allow swiveling 360°
(17)	6	Digital IR Transceiver	Gives 0V on reflective surface, otherwise 5V
(18)	6	Transceiver Cable	3-pin housed wires, male to female
(19)	1	Small Tool Box	F-141, storing mechanical tools and accessories
(20)	1	Tweezers	For handling small components
(21)	1	Spanner	Tool for M3 nut
(22)	1	Hex Key, M3	Tool for M3 socket screw
(23)	1	Hex Key, M4	Tool for M4 socket screw
(24)	1	Screwdriver, Cross	Fasten countersink screws Adjust IR transceiver's sensitivity
(25)	2	Wheel Coupler	Adaptor for installing wheel to motor
(26)	6	Countersink Screw, M3 x 12mm	Countersunk Cross, only for IR transceiver assembly
(27)	6	Nylon Plain Washer, M3	Only for IR transceiver assembly
(28)	6	Flange Nut, M3	Hex, only for IR transceiver assembly
(29)	8	Brass Spacer, M3 x 11mm	Hex, for swivel caster and breadboard assembly
(30)	30	Stainless Steel Screw, M3 x 8mm	Socket Hex, for swivel caster, motor, and breadboard assembly
(31)	40	Stainless Steel Screw, M3 x 12mm	Socket Hex, for car body assembly
(32)	40	Stainless Steel Nut, M3	Hex, use with M3 x 12mm screw
(33)	4	Stainless Steel Screw, M4 x 8mm	Socket Hex, only for wheel assembly

Assembly Instructions

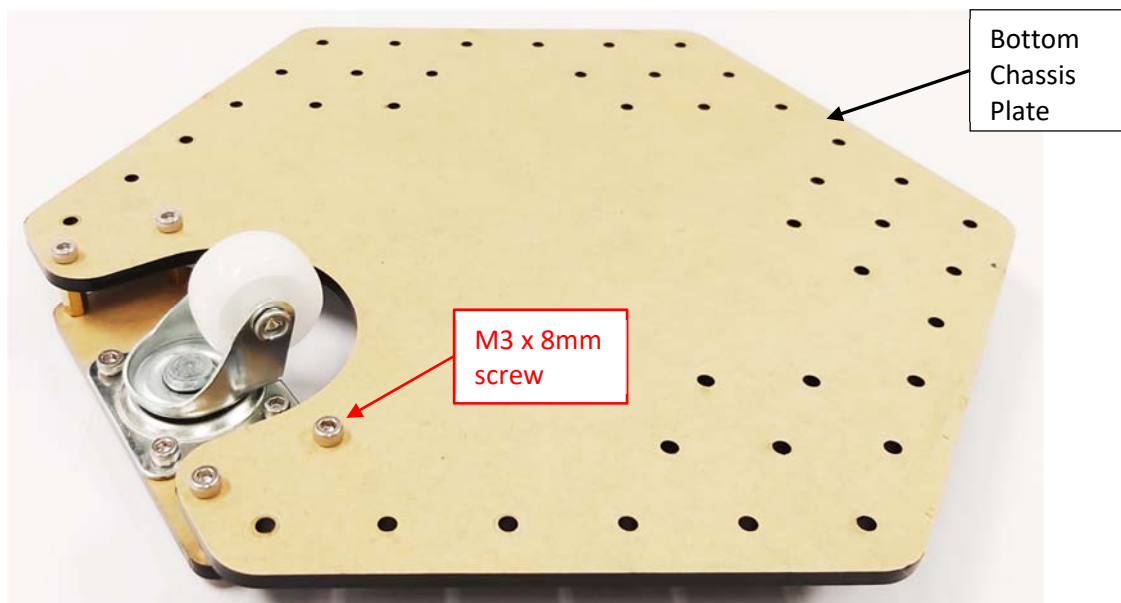
Preparation

- Match the Bill of Materials against the parts in your kit to make sure all parts are present.
- Find a clean workspace on which to assemble your robot.
- Organize tools and parts prior to starting the assembly and separate parts by type to make it easier to gather them during the assembly process.
- Throughout the assembly process, when tightening screws be sure to get them nice and snug but **do not over-tighten** them or you may strip the threads.

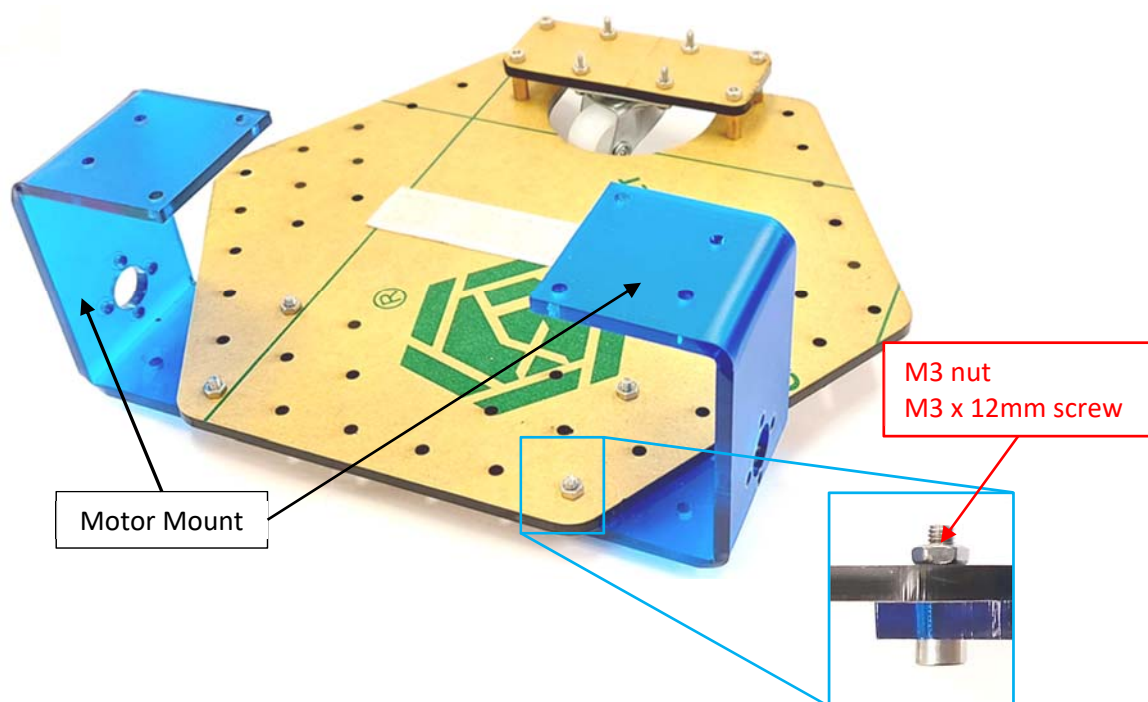
1. Attach the Swivel Caster and Spacers to the Caster Plate



2. Attach the Caster Plate to the Bottom Chassis Plate

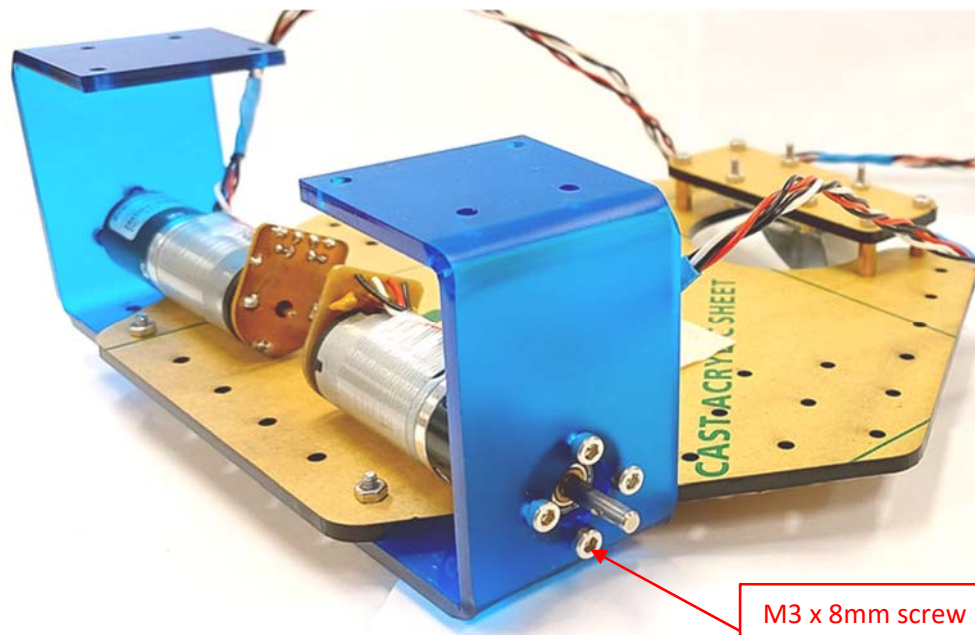


3. Attach two Motor Mounts to Bottom Chassis Plate



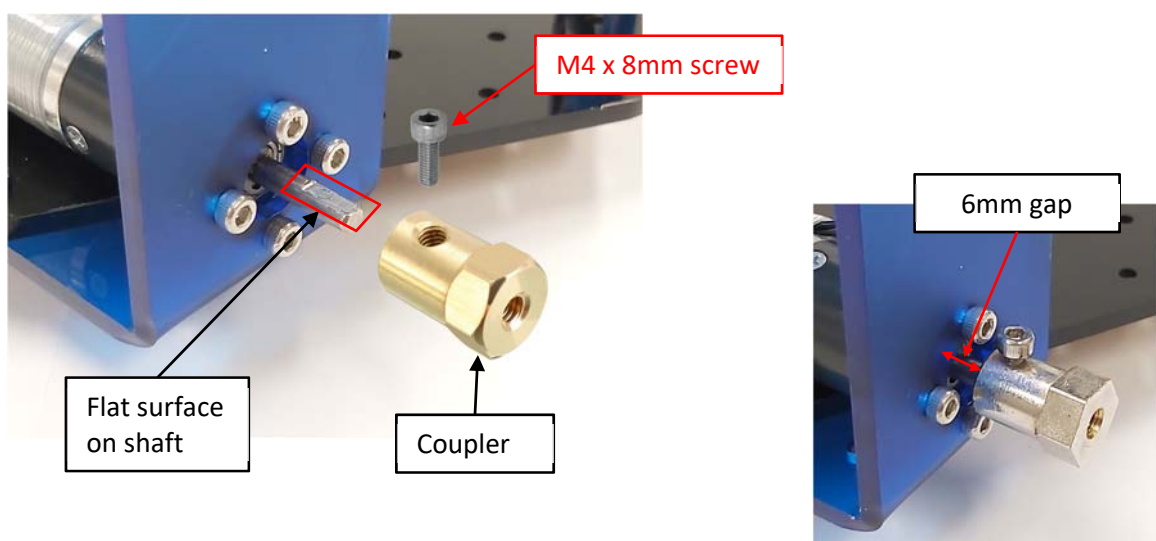
4. Install DC Motors to the Motor Mounts

- a. Use **ONLY M3 x 8mm** screws to install motors. The gearbox would be damaged if long screws are used.



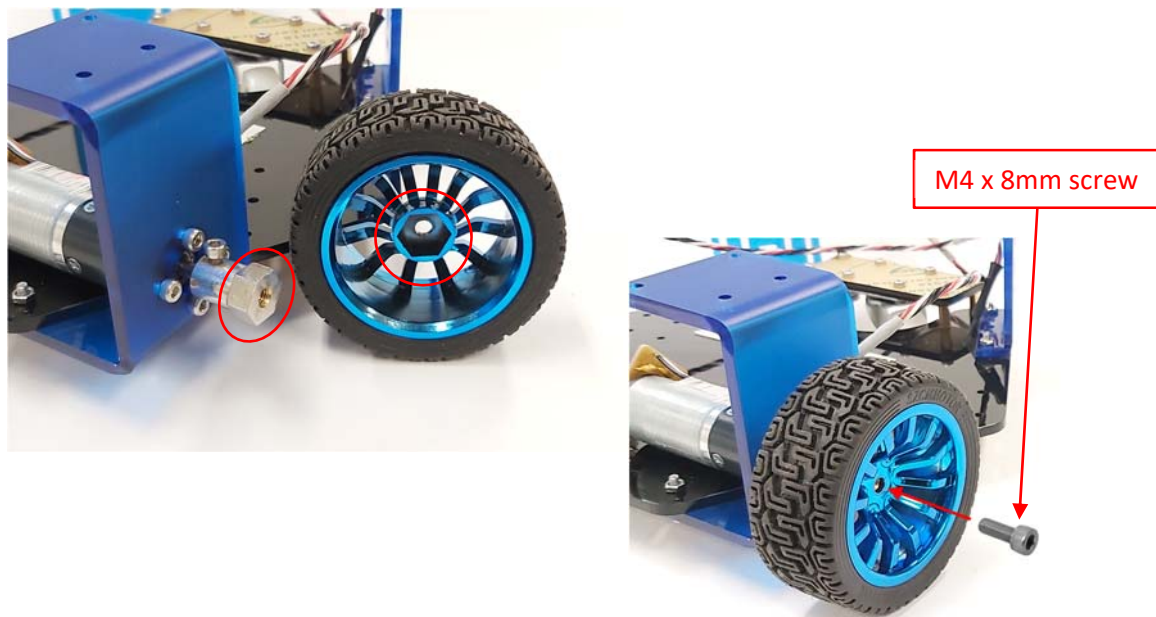
5. Install Wheel Couplers to the Motors

- a. Locate the flat surface of the motor shaft. Install coupler and tighten the screw to the flat surface.
- b. Beware that there should be **6mm gap** between the coupler and the motor mount.

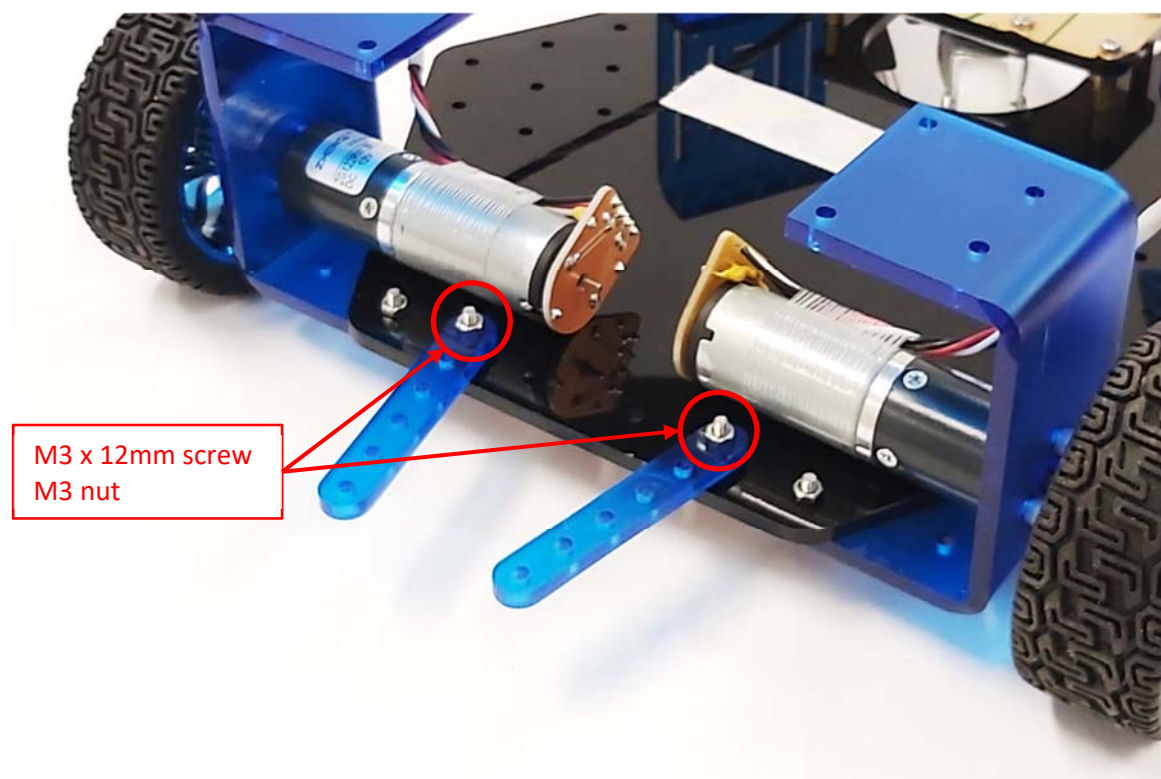


6. Install Wheels to the Couplers

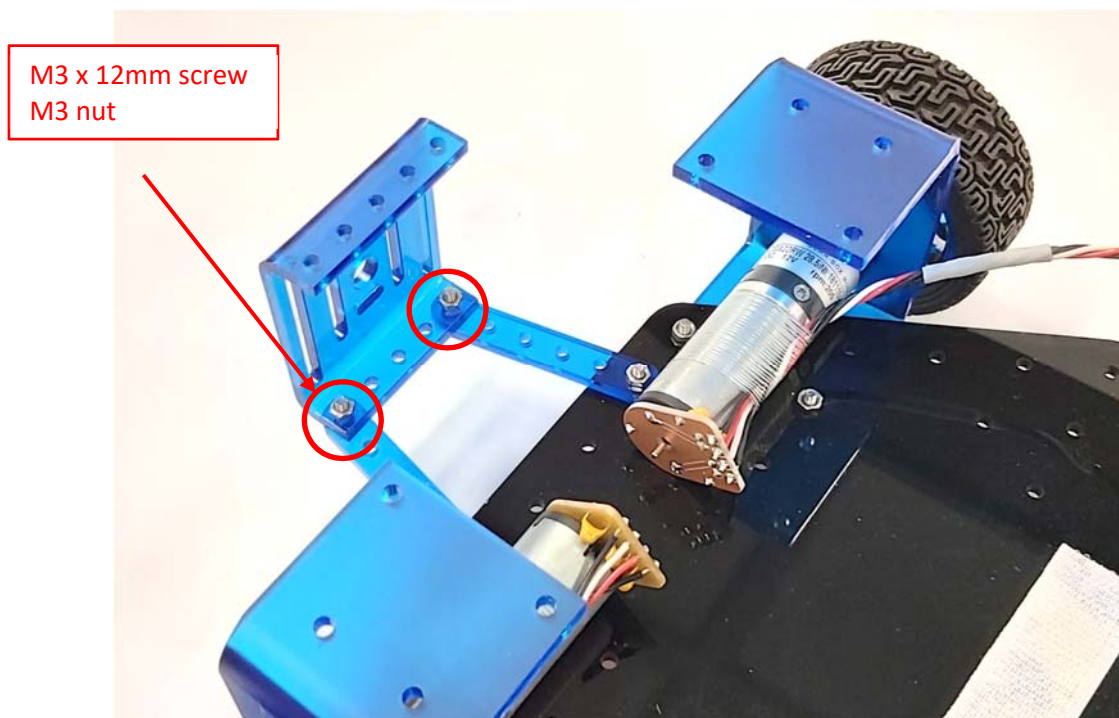
Attach the wheel to the coupler. The hexagonal cavity on the wheel should match the coupler. Use M4 x 8mm screw to tighten the wheel.



7. Add Two 6-hole Bars on Bottom Chassis Plate

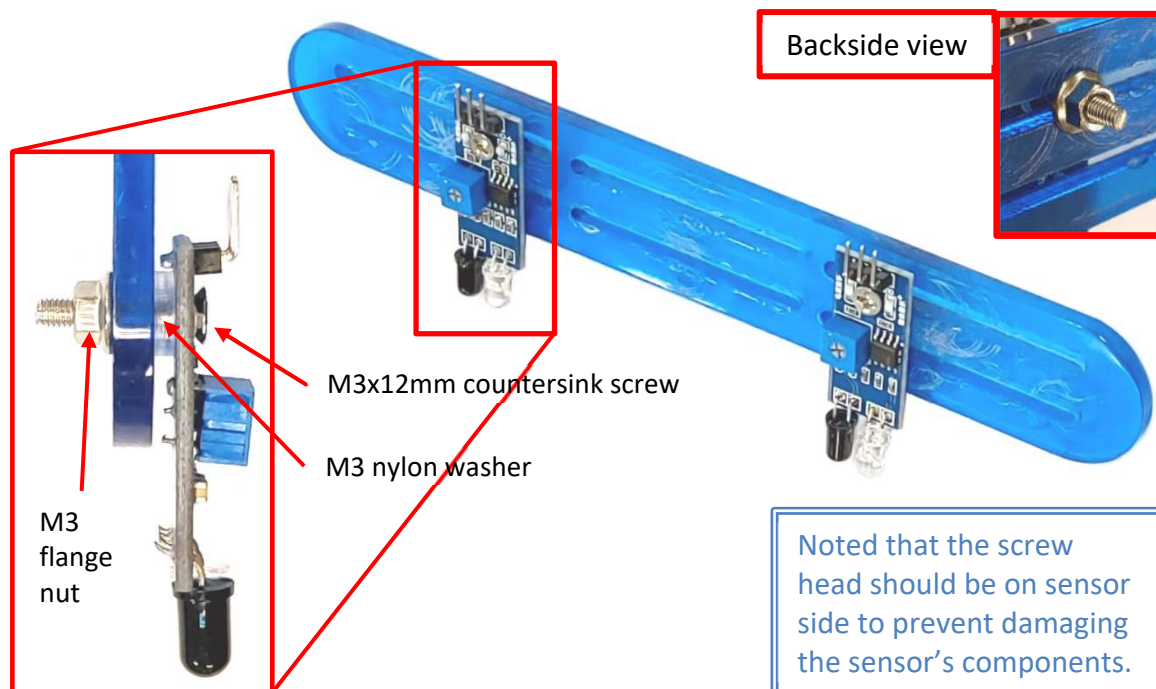


8. Add a Chassis Stand on top of 6-hole Bars



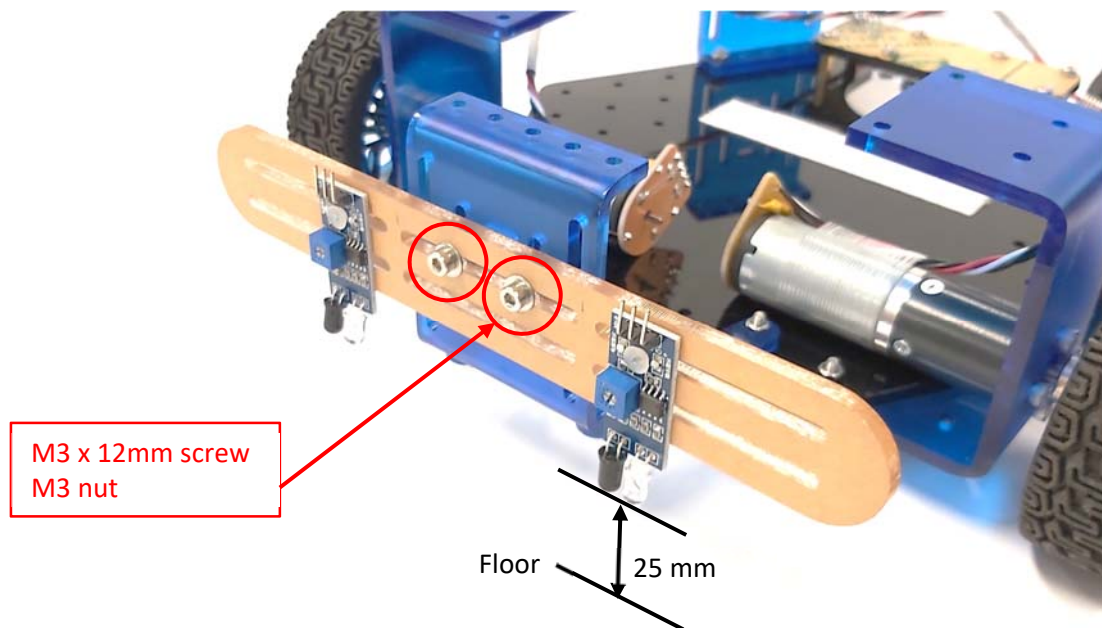
9. Fix Two Digital IR Transceivers on Transceiver Mount Bar

The distance between two transceivers may be varied accordingly to design needs.

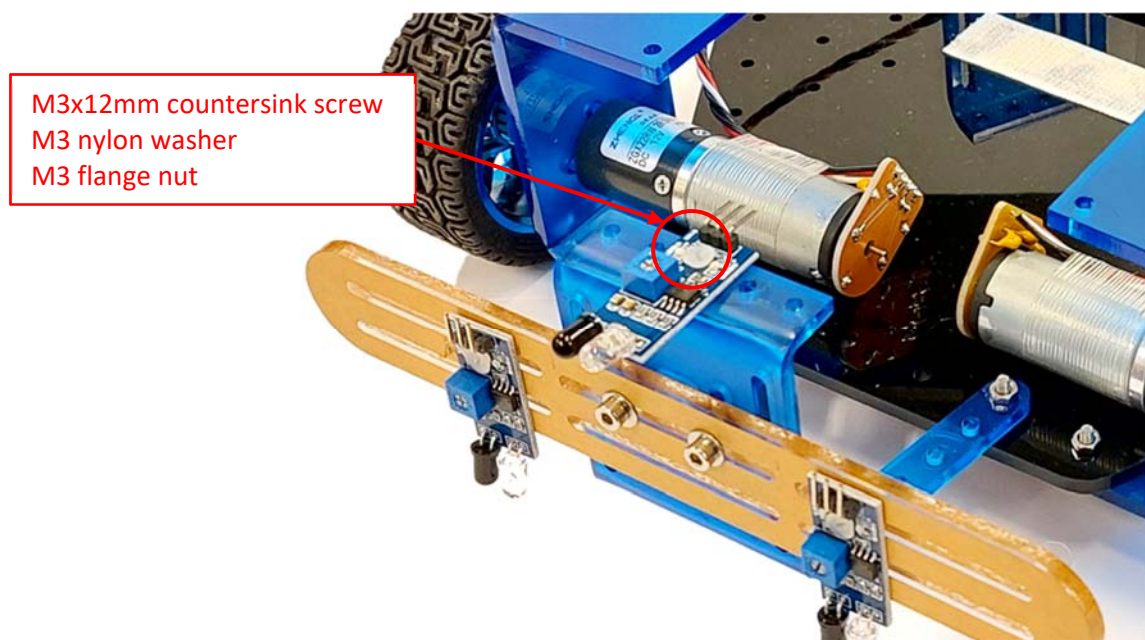


10. Attach Transceiver Mount Bar to front Chassis Stand

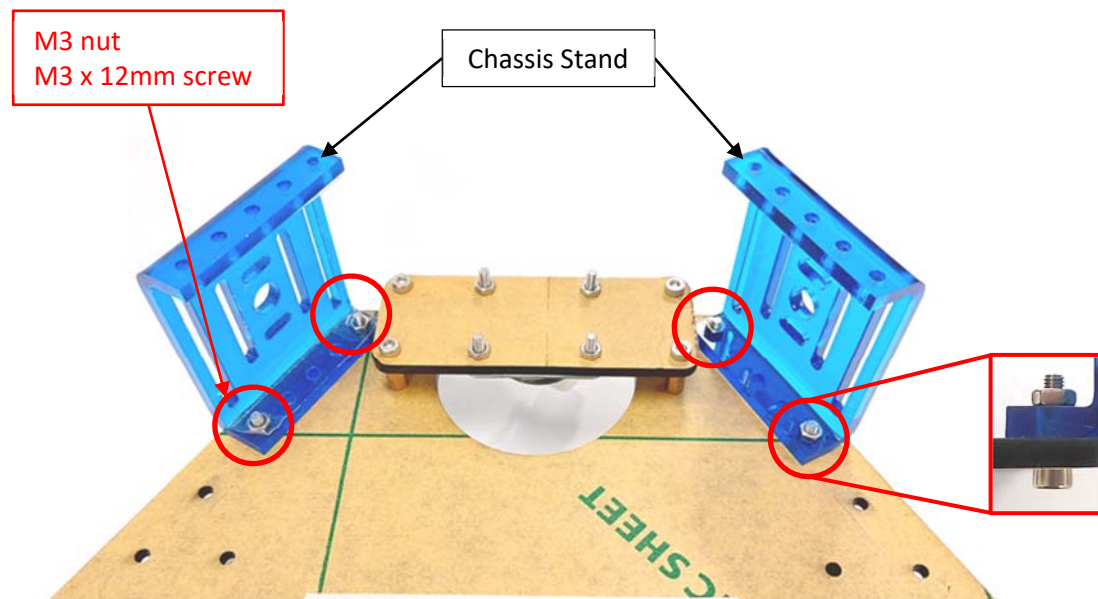
Adjust the vertical position of the transceiver mount bar such that the height for both digital IR transceivers is around **25 mm** from the floor.



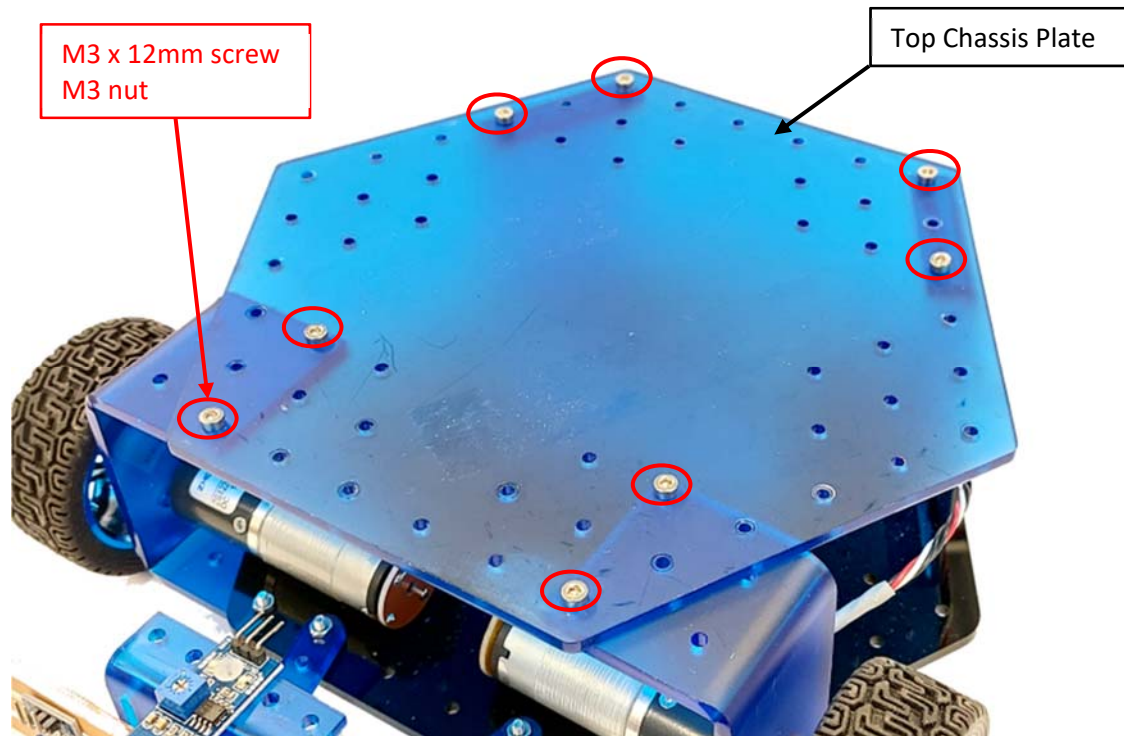
11. Install a Digital IR Transceiver as Bumper Sensor



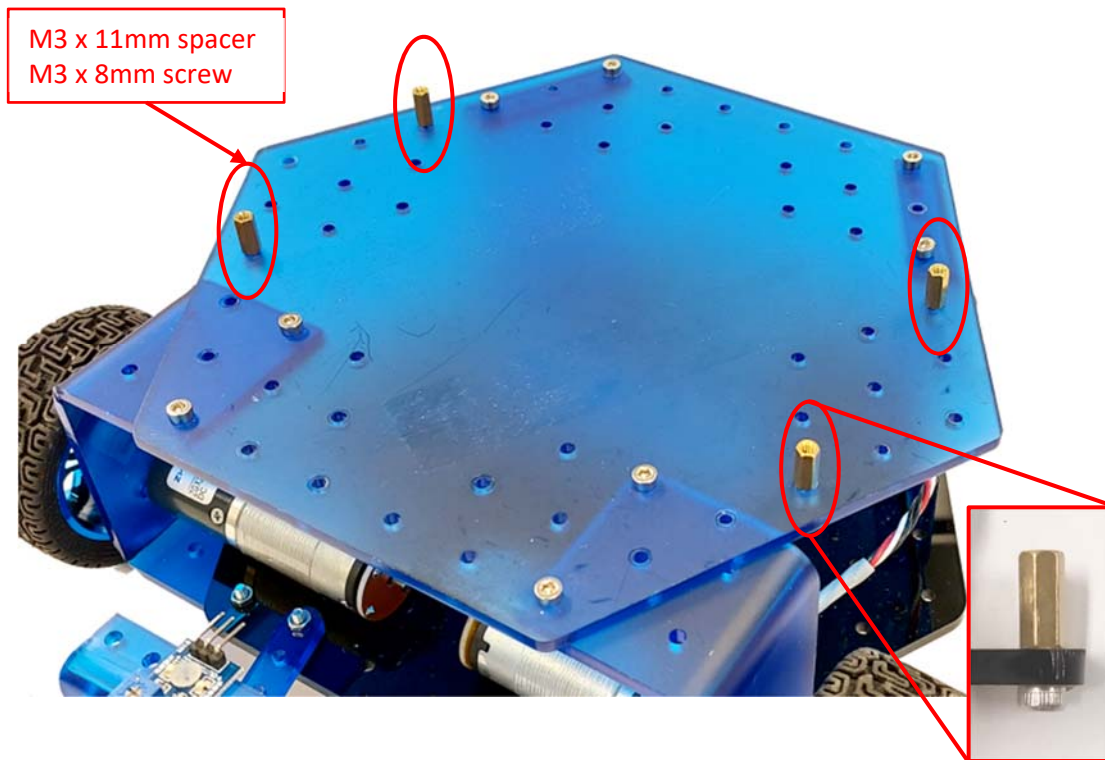
12. Install two Chassis Stands



13. Fix Top Chassis Plate



14. Install 4 Brass Spacers on Top Chassis Plate



15. Fix breadboard on the Spacers

